

Staff Detection and Image-Space Localization for Overhead CCTV

1. Objective

Identify video frames containing staff and estimate each detected staff member's approximate XY position in the source image. Deliver an annotated video, frame-level CSV, and contiguous staff-positive frame ranges.

2. Assumption

A person is staff only when the staff name tag / ID badge is visibly detected on the chest in the current frame. Lack of visible badge evidence yields PERSON for that frame.

3. Pipeline

Video frame	Person detection	Upper-body crop	Badge detection	STAFF / PERSON	Frame + XY
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4. Person Detection

A pretrained YOLO26s detector was fine-tuned using manually reviewed overhead CCTV person annotations. The adaptation targets top-down and fisheye views, seated people, and partial occlusion. The detector proposes person boxes for each frame.

5. Staff Detection

A separate fine-tuned YOLO26n model detects the single `staff_tag` class. It runs on upper-body crops from detected people. Badge evidence from the **current frame** determines STAFF; boxes are drawn only for current detections.

6. Tracking

Tracking provides short-term IDs and bounding-box continuity for visualization. It does not propagate staff identity. This reduces false staff transfer during person overlaps and ID switches.

7. XY Coordinates

For the person bounding box (x_1 , y_1 , x_2 , y_2), image-space location is its bottom-center: $x = (x_1 + x_2) / 2$; $y = y_2$. The origin is the image's top-left. No physical or world-coordinate calibration is assumed.

8. Output

The pipeline writes `outputs/final/staff_detection.mp4`, a per-person detection CSV, and run metadata. `make_submission_results.py` writes staff-only detections, one CSV row per processed frame with staff count and XY coordinates, and staff-positive frame ranges under `outputs/submission/`.

9. Limitations

The models were adapted with limited overhead CCTV data. Generalization to unseen cameras may vary. Tiny, blurred, turned-away, or fully occluded badges can be missed. XY is an image-space approximation, not a world position.

10. Interview Usage

From the project root, install requirements and run `python main.py --video new_test_video.mp4`. Then run `python make_submission_results.py`. Review the annotated video and `outputs/submission/staff_frames_xy.csv`.